

DECENT COACHING CLASSES

X SB

Mind Your Math - II

Subject: Mathematics (Part I & Part II)

Time: 1 Hour

Total Marks: 30

Note: (i) All questions are compulsory. (ii) Use of calculator is not allowed. (iii) Figures to the right indicate full marks.

Q.1 (A) Choose the correct alternative. [4]

- For simultaneous equations in variables x and y : $D = 6$, $D_x = 12$, $D_y = -18$, then $(x, y) =$ (1)
(a) (2, -3) (b) (-2, 3) (c) (3, -2) (d) (2, 3)
- Which of the following is a quadratic equation? (1)
(a) $x^2 + 2x - 3 = 0$ (b) $2x + 3 = 0$ (c) $x^3 - 1 = 0$ (d) $1/x + 2 = 0$
- First term of an A.P. is 5 and common difference is -2, then its 6th term is (1)
(a) -5 (b) 5 (c) 15 (d) -15
- If the SGST paid on an article is ₹250, then the CGST paid on it is (1)
(a) ₹125 (b) ₹250 (c) ₹500 (d) ₹0

Q.1 (B) Solve the following. [4]

- In $\triangle ABC$, seg $DE \parallel$ seg BC . If $AD = 3$, $DB = 6$, $AE = 4$, find EC . (2)
- The sides of a triangle are 9 cm, 12 cm and 15 cm. State with reason whether the triangle is a right-angled triangle. (2)

Q.2 (A) Complete the following activities. (Any 2 of 3) [4]

- Complete the activity to solve the simultaneous equations $x + y = 5$ and $2x - y = 1$ by Cramer's Rule. (2)
 $D = \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} = \square$
 $D_x = \begin{vmatrix} 5 & 1 \\ 1 & -1 \end{vmatrix} = \square$
 $D_y = \begin{vmatrix} 1 & 5 \\ 2 & 1 \end{vmatrix} = \square$
 $\therefore x = D_x / D = \square$ $y = D_y / D = \square$
- Complete the activity to determine the nature of roots of $2x^2 + 5x + 3 = 0$. (2)
Comparing with $ax^2 + bx + c = 0$, $a = \square$, $b = \square$, $c = \square$
 $\therefore b^2 - 4ac = \square - \square = \square$
 $\therefore$ roots are real, unequal and $\square$ (rational/irrational)
- Complete the activity to find the sum of the first 10 terms of the A.P. 6, 10, 14, 18, ... (2)
Here $a = \square$, $d = \square$, $n = 10$
 $S_n = n/2 [2a + (n - 1)d]$
 $S_{10} = 10/2 [\square + 9 \times \square] = 5 \times \square = \square$

Q.2 (B) Solve the following questions. (Any 4 of 5) [8]

- A shopkeeper sold an article of taxable value ₹800. The rate of GST on it is 12%. Find the amount of CGST and SGST charged in the tax invoice. (2)
- The areas of two similar triangles are 50 sq.cm and 98 sq.cm. Find the ratio of their corresponding sides. (2)
- Check whether (7, 24, 25) is a Pythagorean triplet. (2)
- Find the volume of a sphere of radius 3 cm. ($\pi = 3.14$) (2)

5. Find the 15th term of the A.P.: 7, 11, 15, 19, ... (2)

Q.3 Solve the following questions. (Any 2 of 3) [6]

1. The sum of the squares of two consecutive natural numbers is 145. Find the numbers. (3)
2. In $\triangle ABC$, $\text{seg } PQ \parallel \text{seg } BC$. If $AP = 4$, $PB = 8$, $AQ = 5$, find AC . (3)
3. The radius of the base of a cylinder is 7 cm and its height is 10 cm. Find its curved surface area and total surface area. ($\pi = 22/7$) (3)

Q.4 Solve the following questions. (Any 1 of 2) [4]

1. The sum of the ages of a father and his son is 45 years. Five years ago, the father's age was 4 times the son's age. Find their present ages. (4)
2. State and prove the Basic Proportionality Theorem (Thales Theorem). (4)

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Q.1 (A) Answers

1. (a) (2, -3) 2. (a) $x^2 + 2x - 3 = 0$ 3. (a) -5 4. (b) ₹250

Q.1 (B) Answers

1. $AD/DB = AE/EC \Rightarrow 3/6 = 4/EC \Rightarrow EC = 8$
2. $9^2 + 12^2 = 81 + 144 = 225 = 15^2$. Since the square of the largest side equals the sum of squares of the other two sides, the triangle is a right-angled triangle (converse of Pythagoras theorem).

Q.2 (A) Answers

1. $D = (1)(-1) - (1)(2) = -3$; $D_x = (5)(-1) - (1)(1) = -6$; $D_y = (1)(1) - (5)(2) = -9$; $x = D_x/D = 2$, $y = D_y/D = 3$
2. $a = 2$, $b = 5$, $c = 3$; $b^2 - 4ac = 25 - 24 = 1$; since $\Delta > 0$ and a perfect square, roots are real, unequal and rational
3. $a = 6$, $d = 4$, $n = 10$; $S_{10} = 5[12 + 36] = 5 \times 48 = 240$

Q.2 (B) Answers

1. $GST = 12\% \text{ of } 800 = ₹96$; $CGST = SGST = ₹48 \text{ each}$
2. Ratio of areas = $50 : 98 = 25 : 49 \Rightarrow \text{ratio of sides} = \sqrt{25} : \sqrt{49} = 5 : 7$
3. $7^2 + 24^2 = 49 + 576 = 625 = 25^2$ — Yes, (7, 24, 25) is a Pythagorean triplet
4. $V = \frac{4}{3} \pi r^3 = \frac{4}{3} \times 3.14 \times 27 = 113.04 \text{ cm}^3$
5. $a = 7$, $d = 4$; $t_{15} = a + 14d = 7 + 56 = 63$

Q.3 Answers

1. Let the numbers be x , $x+1$. $x^2 + (x+1)^2 = 145 \Rightarrow 2x^2 + 2x - 144 = 0 \Rightarrow x^2 + x - 72 = 0 \Rightarrow (x+9)(x-8) = 0 \Rightarrow x = 8$ (rejecting -9) $\therefore$ the numbers are 8 and 9
2. $AP/PB = AQ/QC \Rightarrow 4/8 = 5/QC \Rightarrow QC = 10$; $AC = AQ + QC = 5 + 10 = 15$
3. $CSA = 2\pi rh = 2 \times 22/7 \times 7 \times 10 = 440 \text{ cm}^2$; $TSA = 2\pi r(r+h) = 2 \times 22/7 \times 7 \times 17 = 748 \text{ cm}^2$

Q.4 Answers

1. Let father's present age = x , son's present age = y . $x + y = 45$... (i) $x - 5 = 4(y - 5) \Rightarrow x = 4y - 15$... (ii) Substituting in (i): $4y - 15 + y = 45 \Rightarrow 5y = 60 \Rightarrow y = 12$, $x = 33$ $\therefore$ Father's age = 33 years, Son's age = 12 years
2. Statement: If a line is drawn parallel to one side of a triangle intersecting the other two sides in distinct points, then the other two sides are divided in the same ratio. (Standard proof using areas of triangles with the same height, as per textbook, to be reproduced with figure.)